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CS 580: Introduction to Artificial Intelligence

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Analysis Report for Homework Assignment 4

This project presents a program that allows users to test and compare five different ways of building decision trees. The methods included in the program are Random, Least-Values, Most-Values, Gini Index, and Information Gain. The program is created using Python and uses the Tkinter library to show a graphical interface. It works with two CSV files, one for training and testing data. The program builds decision trees using the selected methods and checks how well each performs by making predictions and showing their accuracy.

When the program starts, it shows a welcome screen. Users can choose two decision tree methods from a list on this screen. This part of the program is shown in Figure 1. After picking two methods, users can click a button to build and view both decision trees. The program shows the trees along with the predictions and accuracy for each one. This comparison is shown in Figure 2. Another button allows users to view all five decision trees simultaneously. This makes it easier to see and compare each method's performance. The view of all methods is shown in Figure 3.

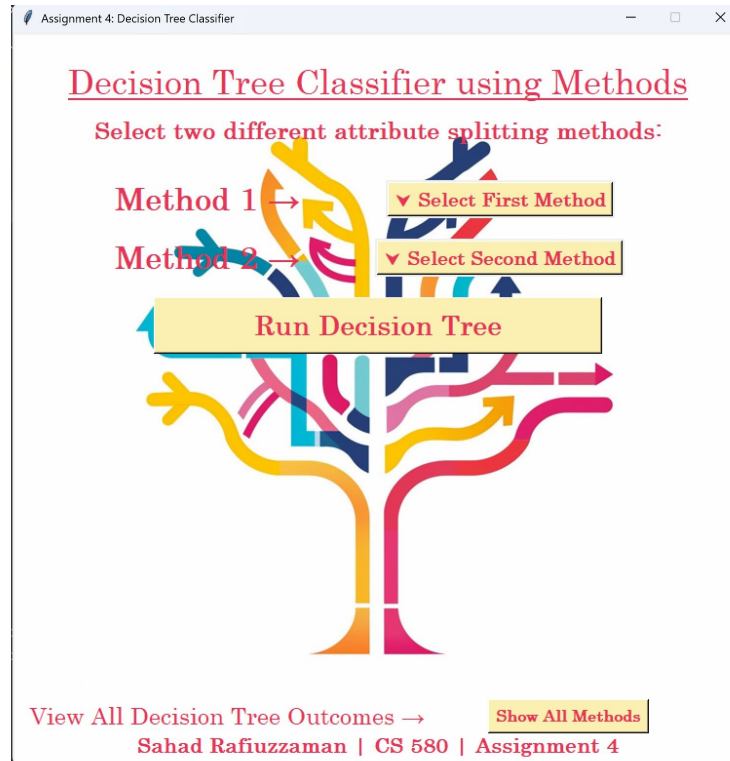


Figure 1: The welcome screen for the Decision Tree Classifier using Methods Project.

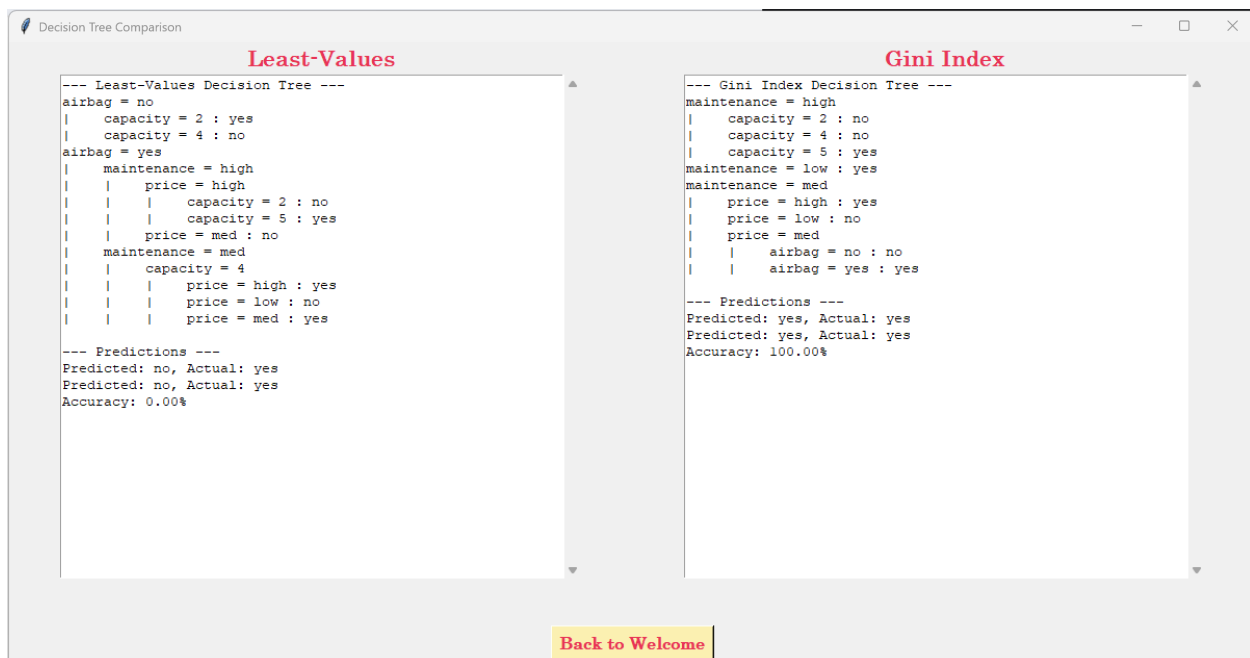


Figure 2: Decision Tree Comparison.

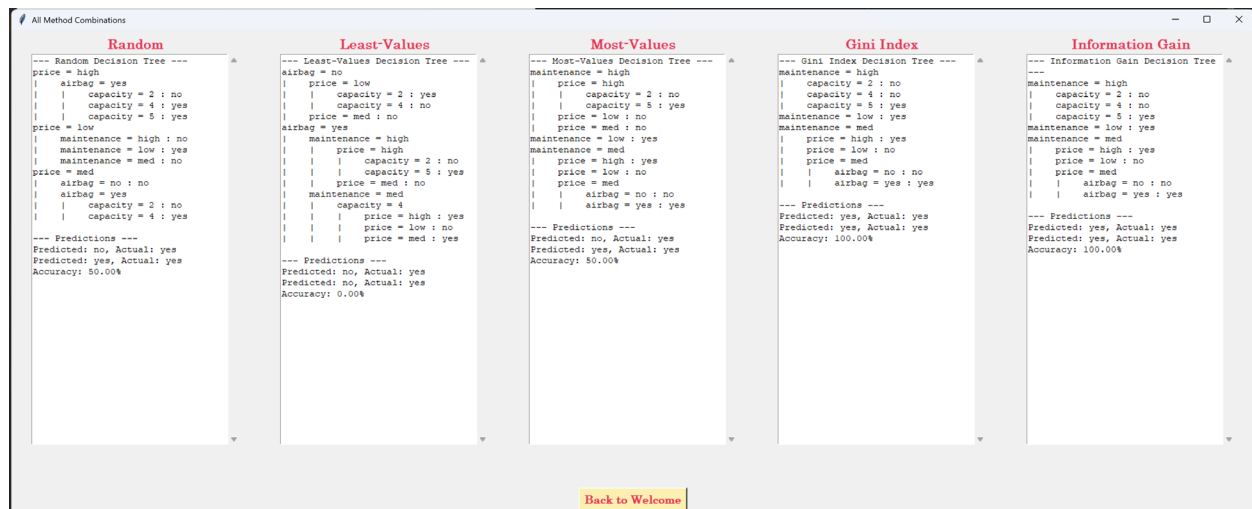


Figure 3: All Methods Option.

Each tree is built by picking the best attribute at each level. The data is then split based on that attribute. The program stops building the tree when all the data in a group has the same label or no more attributes are left to use. Since each method uses a different rule to choose the best attribute, the shapes and paths of the trees are different. This helps show how each method works uniquely.

The program also provides helpful information for the terminal. It prints the size of the training and testing data. It shows the first ten rows from both CSV files. It also shows how many unique values each attribute has. While the program makes predictions, it prints out steps like “Testing attribute airbag, expected value no, actual value no.” These messages show how the tree looks at the data step by step to make a final decision. This helps understand how the tree works and for checking if the logic is correct.

This project makes it easier to learn about decision trees and how different building methods affect their performance. The visual layout helps users see the results more clearly. The terminal output gives extra details for those who want to follow the logic behind the predictions. This program is a helpful tool for learning and exploring decision tree methods.